INCH-POUND

MIL-C-55021/1B
16 April 1991
SUPERSEDING
MIL-C-55021/1A
31 October 1973

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, SHIELDED SINGLES, SHIELDED AND JACKETED SINGLES, TWISTED PAIRS AND TRIPLES, INTERNAL HOOKUP, -40 TO +105°C

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of the specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55021.

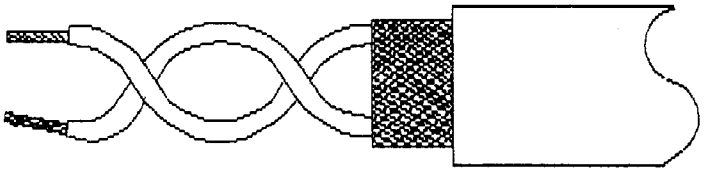


FIGURE 1. Shielded and jacketed cable.

AMSC N/A FSC 6145

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REQUIREMENTS:

Cable construction: The cable shall consist of a shielded singles, shielded and jacketed singles, twisted pairs or triples in accordance with the provisions of MIL-C-55021 (see figure 1 for example).

Type of wire(s): The insulated wires used in constructing the cable shall conform to MIL-W-16878/1 (type B), MIL-W-16878/2 (type C), MIL-W-16878/3 (type D), MIL-W-16878/17 (type BJ), MIL-W-16878/18 (type CJ), or MIL-W-16878/19 (type DJ) as specified (see 6.2 of MIL-C-55021). All wires in the cable shall be of the same type and AWG size.

Length of lay: The length of lay shall be as specified in Table I.

TABLE I. Length of lay.

Wire size range (AWG)	Inches per twist 1/	Twist tolerance 1/
32 to 24	2.0 (50.8)	+ or - 0.5 (12.7)
22 to 18	2.5 (63.5)	+ or - 0.5
16 to 14	3.5 (88.9)	+ or - 0.5
12 to 10	4.5 (114.3)	+ or - 0.5
8 to 6	5.5 (139.7)	+ or - 0.5
4 to 1/0	6.0 (152.4)	+ or - 0.5

1/ Dimensions are in inches. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for information only and are based upon 1 inch = 25.4 mm.

Shield coverage: When specified, (see 1.2.1.5 of MIL-C-55021), the minimum allowable coverage shall be 90 percent, and the angle made by the shielding braid with the longitudinal axis of the twisted conductors shall be between 20 and 40 degrees.

Jacket materials: When specified, (see 1.2.1.5 of MIL-C-55021), jacket materials shall consist of polyvinyl chloride or polyamide.

Jacket wall thickness: The jacket wall thickness using polyvinyl chloride or polyamide shall be in accordance with tables II or III. When a polyamide jacket of greater than .250 inch (6.35 mm) is specified, a polyamide braid in accordance with MIL-C-572 shall be used.

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TABLE II. Polyvinyl chloride jacket thickness.

Diameter over braided shield 1/	Jacket thickness (nominal)	Jacket thickness (minimum)
under .375 (9.53)	.016 (0.41)	.008 (0.21)
.375 and over	.025 (0.64)	.015 (0.38)

1/ When specifying a jacket thickness both unshielded and shielded constructions are applicable.

NOTE: Dimensions are in inches. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for information only and are based upon 1 inch = 25.4 mm.

TABLE III. Polyamide jacket thickness.

Diameter over braided shield 1/	Jacket thickness (nominal)	Jacket thickness (minimum)
under .200 (5.08)	.006 (0.152)	.004 (0.102)
.200 and over	.008 (0.203)	.006 (0.152)

1/ When specifying a jacket thickness both unshielded and shielded constructions are applicable.

NOTE: Dimensions are in inches. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for information only and are based upon 1 inch = 25.4 mm.

Type designation: The type designation shall be in accordance with the provisions of this specification sheet and MIL-C-55021 (see 1.2.1 of MIL-C-55021).

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CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
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Review activities:
Army - CR
DLA - IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-1199-01)

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